Prospective Clinical Validation of a Combinatorial Functional Precision Medicine Platform in Relapsed/Refractory Non-Hodgkin's Lymphoma

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ABSTRACT

PURPOSE Despite initially responding to first-line treatment, many patients with non-Hodgkin's lymphoma (NHL) eventually relapse or are refractory. These patients are empirically subjected to salvage therapies that may not be efficacious. We had previously presented feasibility evidence of an ex vivo functional precision medicine (FPM) platform, quadratic phenotypic optimization platform (QPOP), being potentially useful in identifying alternative therapeutic options for patients with relapsed/refractory (R/R)-NHL. We now present an analysis of the completed prospective study, with clinical concordance and benefit of QPOP in predicting treatment responses of patients with R/R-NHL.

MATERIALS AND METHODS One hundred seventeen patients with R/R B-cell NHL (B-NHL) or natural killer or T-cell NHL (NK/T-NHL) were recruited for QPOP testing. Isolated tumor cells were incubated for 48 hours with drug combinations determined by an orthogonal array composite design. QPOP reports with patient-specific optimal drug therapies were generated. Patients were given off-label treatments according to the clinician's decision. Patient characteristics and responses to treatments after QPOP testing were recorded.

RESULTS Within 126 QPOP cases, 105 (52 B-NHL and 53 NK/T-NHL) were evaluable. QPOP-suggested drug responses were concordant with actual patient outcome, with an overall test accuracy of 74.5%. An overall response rate of 59% was achieved in those prescribed off-label QPOP-guided combinations. In all, 59.3% of QPOP-guided patients had improved response durations compared with their previous treatment. Compared with those who received salvage therapy, QPOP-guided patients also had longer progression-free survival 2 years after treatment.

CONCLUSION There is increasing evidence that genetic factors are not sole determinants of patient drug response and FPM approaches may improve cancer treatment guidance. This study further confirms that advancements in ex vivo combinatorial drug screening platforms like QPOP could complement existing genomic methods in identifying effective treatment options for patients with cancer, especially in R/R cases.

ACCOMPANYING CONTENT

Data Supplement

Protocol

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INTRODUCTION

Non-Hodgkin's lymphoma (NHL) is ranked the 10th most diagnosed malignancy.[1-3] Although patients with this highly heterogeneous hematological malignancy respond well to first-line treatment, approximately 20%-65% eventually relapse or are refractory.[4-7] Salvage therapies often use combination treatments to overcome intratumoral heterogeneity in drug sensitivities, increasing chances of eradicating the entire tumor. However, many combinations are

CONTEXT

Key Objective

The key objective of this study was to investigate whether ex vivo functional precision medicine (FPM) analysis, through a quadratic phenotypic optimization platform (QPOP), identifies patient-specific drug combinations that show efficacy in patients with relapsed/refractory (R/R) non-Hodgkin's lymphoma (NHL).

Knowledge Generated

QPOP was validated to be a robust laboratory assay for ex vivo FPM, with acceptable to excellent coefficient of variance. The assay showed concordance with actual clinical responses, with an overall accuracy of 74.5% at predicting patient treatment outcome. Patients who received top-ranking QPOP-predicted drug combinations had an overall response rate of 59%, with an improved progression-free survival period compared with those who received standard salvage treatment.

Relevance

This study demonstrated that QPOP is a reliable and useful FPM platform that can be implemented as a clinical decision support tool for the choice of combination therapy in patients with R/R NHL.

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based on population-derived drug responses and inter-patient heterogeneity may result in some mismatched patient treatment and poor individual treatment response.[8,9] Furthermore, with increasing lines of treatment, response duration has been shown to progressively shorten.[10,11] Thus, to provide effective treatment for relapsed/refractory (R/R) patients who have exhausted multiple lines of standard-of-care treatment options, reliable approaches to identify effective patient-specific, optimal therapies through ex vivo drug screening must be developed.

The utility of ex vivo drug sensitivity screening platforms, termed as functional precision medicine (FPM), for the guidance of patient treatment is becoming increasingly common.[12] This approach provides alternative therapeutic options, especially for patients who do not harbor targetable mutations.[12,13] It has been explored in many indications, particularly for hematological malignancies.[14-17] We have developed an ex vivo drug combinatorial approach, quadratic phenotypic optimization platform (QPOP), as a potential FPM platform. QPOP leverages on orthogonal array composite design (OACD), which minimizes the number of experimental data points, thereby significantly reducing the amount of tissue required for analysis.[18,19]

QPOP has been applied across different stages of drug development and indications.[18,20-23] Recently, its clinical feasibility has been evaluated in a cohort of patients with R/R-NHL.[24] Through this current study, we have expanded upon the initial feasibility data by applying QPOP to a larger prospective cohort of patients with R/R-NHL to evaluate its clinical concordance and clinical benefit as a clinical decision support (CDS) tool. This study provides clinical evidence that QPOP may be a useful tool to aid treatment decision in R/R-NHL.

MATERIALS AND METHODS

Study Design

This completed study is an extension of the previously reported prospective study evaluating QPOP as a combinatorial FPM platform in the setting of R/R-NHL.[24] Following the previous cutoff date of December 1, 2021, additional 51 patient samples (25 B-cell NHL [B-NHL], 26 natural killer or T-cell NHL [NK/T-NHL]) were collected. After excluding nonviable or non-NHL samples and conducting data quality assessments (Z-score >0.5, strictly standardized mean differences >3), 105 evaluable QPOP cases were obtained. The first and last patients were recruited on November 1, 2017, and April 25, 2023, respectively. To allow a minimal of 6 months of follow-up after QPOP testing, November 1, 2023, was selected as the censor date. Patients were recruited from two Singapore tertiary oncology centers, and all recruitment processes were conducted with institutional review board (IRB) approvals (IRB: H-18-032, 2015/00176 and 2019/00271) and informed consent. Criteria for patient selection, sample collection, and treatment plan following QPOP testing were as previously described.[24] Patient treatment allocation was nonrandomized and dependent on clinician's decision after evaluating the QPOP reports. Patients treated with drugs listed within the top three QPOP combinations were classified as QPOP-guided, whereas patients treated according to clinician preference with drugs outside of the top ranking QPOP combinations were classified as clinician-guided. Patient identities were kept anonymous to individuals outside the research group by using QPOP case identifiers. Cost of care was not covered by the trial. Standard treatment costs were paid for by patients or through insurance. Off-label treatment was covered under compassionate use or by patients themselves.

Sample Processing and Drug Combination Testing

As previously described,[24] drug concentrations used in QPOP testing were fine-tuned over time based on cumulative single-drug sensitivity data and restrained to below 10% maximum serum concentration. The patient specimens were immediately processed and treated with drug combinations derived through an OACD, previously established by Xu et al,[19] for optimal drug combination interrogation. Cell viability was measured using CellTiter-Glo Luminescent Assay (Promega, Madison, WI) 48 hours after treatment, according to manufacturer's instructions. Readings were normalized to untreated controls and analyzed with QPOP. Procedures are further detailed in the Data Supplement.

QPOP Analysis Pipeline

Similar to previous QPOP studies,[18,22,24] normalized cell viability (NCV) values served as input data and were fitted into a second-order quadratic series using Optim.AI software (KYAN Technologies, Singapore). Through model fitting, QPOP predicts the viability values for all other possible drug permutations. For clinical applications, one-, two-, and three-drug combinations were evaluated and the ex vivo cell killing score (ECKS) was calculated by subtracting the viability value from 1.

Patient Response Assessment and Clinical Concordance Analysis

Patients treated with standard salvage regimens were followed up every 2-3 weeks, whereas those treated with novel regimens were monitored weekly. Radiologic assessments were conducted every 6-8 weeks. Patient response to treatment was evaluated on the basis of the Lugano criteria.[25] Patients who had received at least one drug within the QPOP drug panel with corresponding clinical outcomes were evaluated for the concordance analysis. The 47 evaluable cases consisted of 27 patients whose treatment was guided by QPOP, whereas the other 20 were treated based on clinicians' choice. The receiver operating characteristic (ROC) curve was plotted using NCVs for each clinical outcome to identify an NCV cutoff that stratifies clinical responders (complete response [CR] and partial response [PR]) from nonresponders (stable and partial disease). On the basis of this identified cutoff, the following parameters were calculated (Table 1).

Statistical Analysis

Kaplan-Meier analysis was used to compare progression-free survival (PFS) of enrolled QPOP patients. Candidates who had no response assessment or underwent clinical trials for immunotherapies or allogenic stem cell transplant were excluded. All statistical analyses were conducted using Prism 10 (GraphPad). Unless otherwise stated, a P value threshold <.05 was applied for statistical significance.

RESULTS

Characteristics of Patients Enrolled for QPOP Testing

To evaluate the potential of QPOP as a CDS tool, 126 tumor biopsies were collected from 117 patients with R/R-NHL. After data processing, 105 QPOP generated reports were evaluable (Fig 1). The overall QPOP workflow is outlined in Figure 2A. Median sample-to-report time was 6 days (Data Supplement, Fig S1). Enrolled patients were broadly classified into B-NHL or NK/T-NHL. Expectedly, diffuse large B-cell lymphoma (DLBCL) and peripheral T-cell lymphoma (PTCL) were the most common subtypes, respectively[26-28] (Figs 2B and 2C). Overall, the number of evaluable cases for B-NHL (n = 52) and NK/T-NHL (n = 53) were comparable. The median age of enrolled patients was 58 years (IQR, 48-65 years) and the median follow-up period was 5.6 months (IQR, 2.36-16.28 months) since QPOP testing (Table 2). Patient characteristics and demographics are further detailed in Table 2.

Analytical Validation of QPOP

To implement QPOP as a laboratory developed test conducted within clinically certified premises, analytical validation (intra- and interassay reproducibility and intermediate precision parameters) was carried out, as detailed in the Data Supplement, to demonstrate the assay's precision, accuracy, and robustness across heterogeneous patient samples (Data Supplement, Fig S2).

With a 20% coefficient of variance (CV) threshold,[29] the results showed acceptable to excellent CV on at least 80% of the drug combinations tested, for intra- and interassay reproducibility and intermediate precision parameters (Data Supplement, Figs S2A and S2B). From the array of drug combinations tested, three representative combinations

TABLE 1. Calculations of Sensitivity, Specificity, PPV, NPV and Accuracy Parameters for QPOP Clinical Concordance Analysis

Responder *v* Non-Responder	Clinical Responder	Clinical Non-Responder	Parameter
QPOP responder	TP	FP	PPV = TP / (TP + FP)
QPOP non-responder	FN	TN	NPV = TN / (TN + FN)
Parameter	Sensitivity = TP / (TP + FN)	Specificity = TN / (TN + FP)	Accuracy = (TP + TN) / Total N

Abbreviations: FN, false negative; FP, false positive; NPV, negative predictive value; PPV, positive predictive value; QPOP, quadratic phenotypic optimization platform; TN, true negative; TP, true positive.

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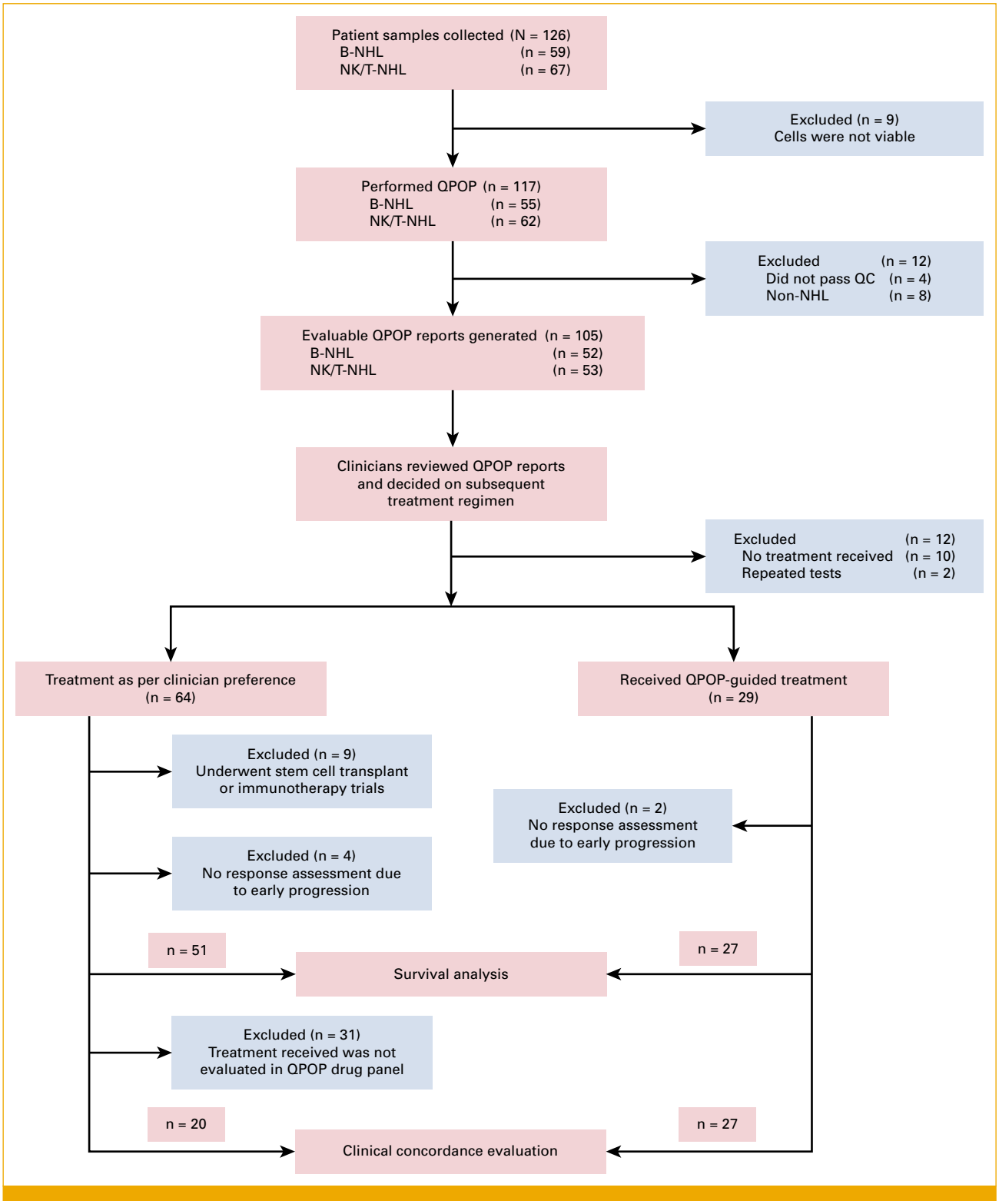


FIG 1. CONSORT diagram of the completed NHL-QPOP study. B-NHL, B-cell NHL; NHL, non-Hodgkin's lymphoma; NK/T-NHL, natural killer or T-cell NHL; QC, quality check; QPOP, quadratic phenotypic optimization platform.

(3, 9, and 7 corresponding to three-, four-, and five-drug combinations, respectively) were highlighted to demonstrate the reproducibility for both intra-assay and interassay parameters (Data Supplement, Figs S2C-S2F). No significant difference was observed for combinations 3 and 7, but significant differences were observed for 9, which could be more volatile due to the low viability. Overall, the analytical validation results highlight the ability of the laboratory and assay to generate reports with robust QPOP results on patient samples.

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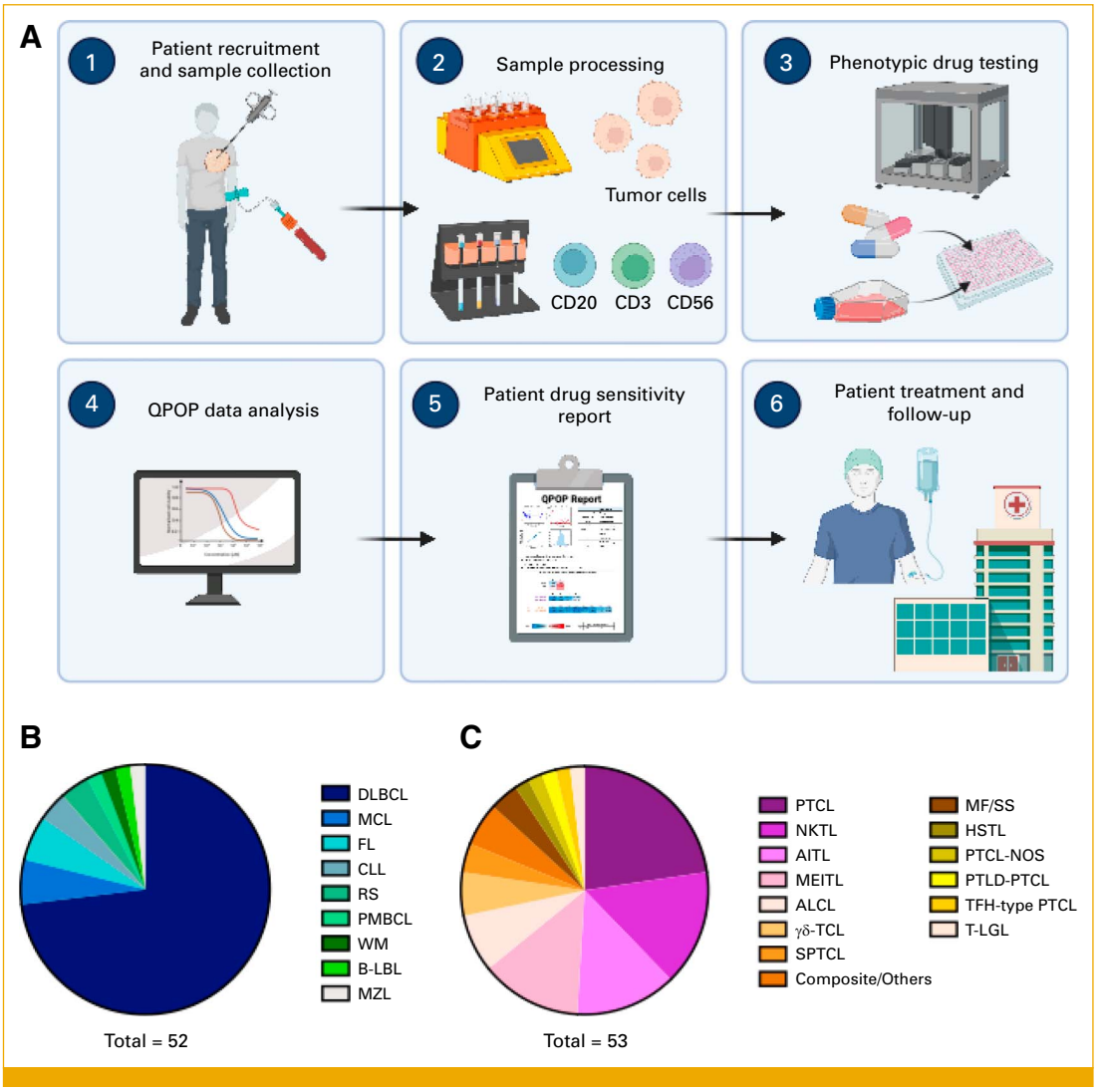


FIG 2. Overview of patient characteristics and QPOP testing workflow. (A) Schematic diagram outlining the QPOP testing process. Briefly, tissue biopsies or blood aspirates obtained from the patient are processed and enriched for tumor cells. Isolated tumor cells were incubated with various drug combinations for 48 hours. Cell viability was determined via CellTiter-Glo. Normalized viability scores were used as input data for QPOP analysis. Top ranking combinations were consolidated in a patient report for clinicians. Patients were subsequently treated and followed up. (B) B-NHL and (C) NK/T-NHL subtype proportions among the evaluable QPOP cases. AITL, angioimmunoblastic T-cell lymphoma; ALCL, anaplastic large cell lymphoma; B-LBL, B-cell lymphoblastic lymphoma; B-NHL, B-cell NHL; CLL, chronic lymphocytic leukemia; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; HSTL, hepatosplenic T-cell lymphoma; MCL, mantle cell lymphoma; MEITL, monomorphic epitheliotropic intestinal T-cell lymphoma; MF/SS, mycosis fungoides/Sézary syndrome; MZL, marginal zone lymphoma; NHL, non-Hodgkin's lymphoma; NK/T-NHL, natural killer or T-cell NHL; NKTL, natural killer T-cell lymphoma; NOS, not otherwise specified; PMBCL, primary mediastinal large B-cell lymphoma; PTCL, peripheral T-cell lymphoma; QPOP, quadratic phenotypic optimisation platform; RS, Richter's syndrome; SPTCL, subcutaneous panniculitis-like T-cell lymphoma; TFH, T-follicular helper; T-LGL, T-large granular lymphocytic leukemia; WM, Waldenstrom macroglobulinemia; γδ-TCL, gamma-delta T-cell lymphoma.

Summary of Evaluable Cases and Drug Combination Analyses

To predict effective drug combinations for each patient, QPOP was carried out on the collected biopsies. The selected drug panels and concentrations of each drug used were optimized as previously described.[24] The drugs used for QPOP testing of individual patient samples and the corresponding patient features are displayed in the Data Supplement (Figs S3A and S3B).

For patients recruited after the initial reported study,[24] dosages of several drugs, such as bortezomib and copanlisib and venetoclax, were adjusted (Data Supplement, Figs

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TABLE 2. Summary of Patient Characteristics

Characteristic	Overall Cohort (n = 105)	B-NHL (n = 52)	NK/T-NHL (n = 53)
Age at diagnosis, year			
Median (IQR)	58 (48-65)	60 (53.25-66.75)	56 (45-64)
Gender, No. (%)			
Male	59 (56.2)	26 (50.0)	33 (62.3)
Female	46 (43.8)	26 (50.0)	20 (37.7)
Stage at diagnosis, No. (%)			
I	3 (2.8)	0	3 (5.7)
II	11 (10.4)	5 (9.6)	6 (11.3)
III	16 (16.0)	8 (15.4)	8 (15.1)
IV	68 (64.2)	37 (71.2)	31 (58.5)
Unknown	7 (6.6)	2 (3.8)	5 (9.4)
Lines of treatment befoe QPOP testing, No. (%)			
1	34 (32.4)	15 (28.9)	19 (35.9)
2	34 (32.4)	14 (26.9)	20 (37.7)
≥3	37 (35.2)	23 (44.2)	14 (26.4)
Follow up period since diagnosis, months			
Median (IQR)	27.85 (14.21-55.06)	31.98 (16.67-75.10)	23.25 (12.97-38.64)
Follow-up period since QPOP testing, months			
Median (IQR)	5.6 (2.36-16.28)	6.58 (3.52-18.95)	5.60 (2.36-17.62)

NOTE. One hundred five patients were evaluated for this study, including six patients with repeated biopsies at different time points or multiple tumor sites.
Abbreviations: B-NHL, B-cell NHL; NHL, non-Hodgkin's lymphoma; NK/T-NHL, natural killer/T-cell NHL; QPOP, quadratic phenotypic optimization platform.

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S3A and S3B). Thus, correlation of single-drug sensitivity to QPOP-derived two-drug combination rankings was reassessed. Strong positive and significant associations were observed for targeted therapies bortezomib and venetoclax, whereas weak associations were observed for chemotherapies oxaliplatin and bendamustine. These trends did not apply for copanlisib and vinorelbine (Data Supplement, Fig S4). Additionally, cohort-wide analyses of recurring two-drug combinations revealed that copanlisib- and romidepsin-based combinations remained as the most frequently recurring doublets in patients with B-NHL (Fig 3A) and NK/T-NHL (Fig 3B), respectively. Overall, the results demonstrated that single-drug dose responses may not fully predict the efficacy of drug combinations. Ex vivo combinatorial FPM approaches like QPOP can potentially identify patient-specific, optimal drug combinations.

Case Studies: Responders to QPOP-Guided Treatments

During this 6-year study, 29 patients who exhausted standard-of-care options had received QPOP-guided drug combinations. This was decided by clinicians after evaluating the QPOP results and other external factors, including patient status. An example of a complete responder following QPOP-guided treatment was a 36-year-old male initially diagnosed with PTCL-T-follicular helper (TFH) phenotype (Fig 4A). He subsequently relapsed and was rediagnosed with composite PTCL-TFH, Epstein-Barr virus–negative DLBCL. When his disease relapsed after third-line treatment, he was enrolled into the study. QPOP analysis had predicted that single drug bendamustine would outperform other standard therapies (Fig 4B). Although bendamustine is not approved for PTCL, several studies demonstrated its efficacy in TCL, including the BENTLY trial where 20% of patients achieved CR.[30–32] Furthermore, its combination with romidepsin was explored in a small study of four patients with R/R PTCL, with two complete responders.[33] Thus, he was treated with six cycles of bendamustine, romidepsin, and brentuximab. A positron emission tomography scan conducted 1 month after treatment showed disappearance of the tumor and was compatible with CR (Fig 4C). By the censor date, the patient had been progression-free for 681 days and continues to have sustained response.

A second example of a complete responder to QPOP-guided therapy was a 75-year-old male diagnosed with PTCL. This patient received six cycles of doxorubicin, cyclophosphamide, vincristine, prednisone and was enrolled into the QPOP study upon disease relapse (Fig 4D). QPOP predicted that combination ifosfamide, venetoclax, and dexamethasone could be more effective than other standard therapies (Fig 4E). Although this combination has yet to be formally tested in PTCL, the combined usage of ifosfamide and venetoclax has been trialed in a phase I (VICER) study where 69% of patients treated for R/R DLBCL had CR.[34] Thus, the

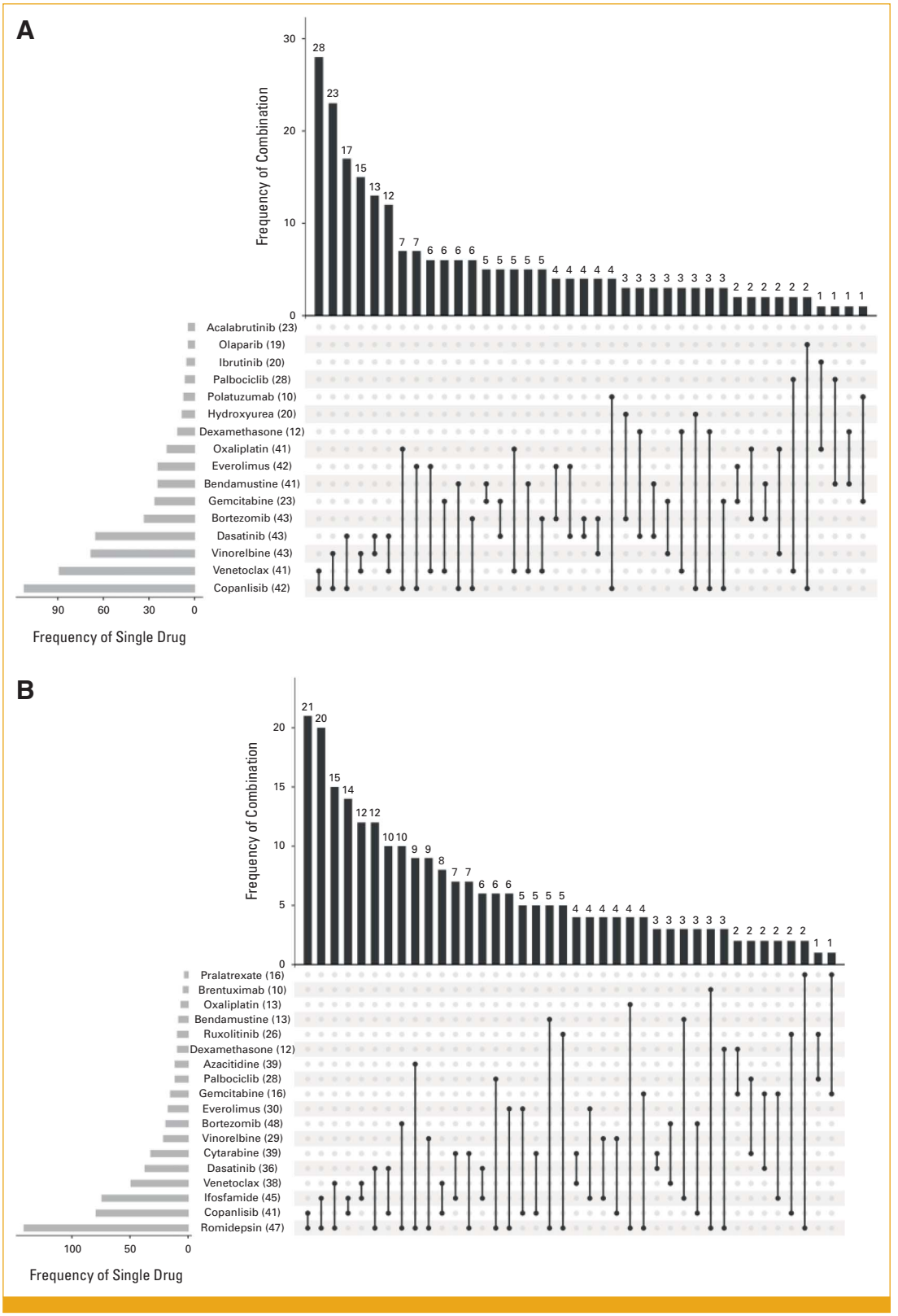


FIG 3. Top ranking two-drug combinations in B- and NK/T-NHL cohorts. UpSet plot depicting the number of times a pair of drugs appeared within the top 10 two-drug combination (top bar graph) in the (A) B-NHL (n = 43) and (B) NK/T-NHL (n = 48) cohorts. A drug pair is denoted with a line connecting two dots shown at the bottom of the plot. The bar graph frequency of single drug on the left represents the number of times a drug has appeared within the top 10 two-drug combination. The number in parentheses represents the number of cases in which the drug has been used for ex vivo testing. B-NHL, B-cell NHL; NHL, non-Hodgkin's lymphoma; NK/T-NHL, natural killer or T-cell NHL.

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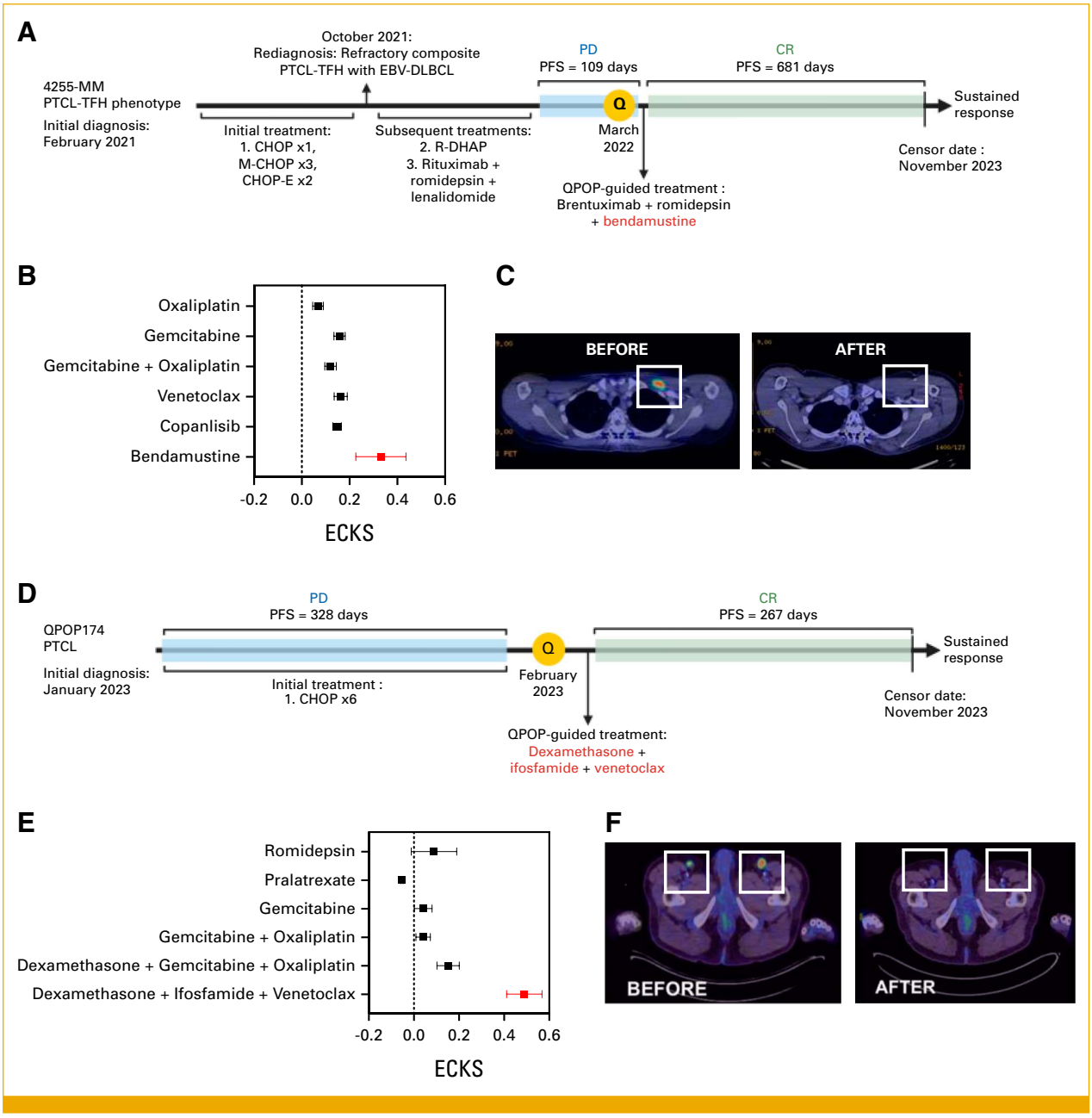


FIG 4. Complete responders to QPOP-guided treatments. (A) Treatment timeline of case 4255-MM, a patient initially diagnosed with PTCL-TFH and subsequently rediagnosed with composite PTCL-TFH and DLBCL. Yellow circle denotes the time at which QPOP testing was conducted (March 2022). Blue and green bars represent the response period of the treatment just before and after QPOP-guided treatment, respectively. (B) ECKS (mean ± SD) of single drug bendamustine (in red) in comparison with other lines of salvage therapies. (C) PET scan before (left) and after (right) QPOP-guided brentuximab, romidepsin, and bendamustine treatment. (D) Treatment timeline of case QPOP174, a patient diagnosed with PTCL. Yellow circle denotes the time at which QPOP testing was conducted (February 2023). Blue and green bars represent the response period of the treatment just before and after QPOP-guided treatment, respectively. (E) ECKS (mean ± SD) of combination dexamethasone, ifosfamide, and venetoclax (in red) in comparison with other standard salvage combination treatments. (F) PET scan before (left) and after (right) treatment with dexamethasone, ifosfamide, and ventoclax. CHOP, cyclophosphamide + doxorubicin + vincristine + prednisone; CR, complete response; DLBCL, diffuse large B-cell lymphoma; E, etoposide; EBV, Epstein-Barr virus; ECKS, ex vivo cell killing score; M, methotrexate; PD, progressive disease; PET, positron emission tomography; PFS, progression-free survival; PTCL, peripheral T-cell lymphoma; QPOP, quadratic phenotypic optimisation platform; R-DHAP; rituximab + dexamethasone + cytarabine + cisplatin; SD, stable disease; TFH, T-follicular helper.

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patient was prescribed this off-label, QPOP-guided drug combination, where he achieved CR (Fig 4F) and a PFS of 267 days with sustained response.

QPOP-Predicted Results Are Concordant With Clinical Responses and QPOP-Guided Treatments Improve Patient Outcomes

Although individualized responses to QPOP-guided treatment highlight the potential of FPM in improving patient-specific treatment, analysis of QPOP's overall clinical concordance is necessary to determine its broad clinical utility as an FPM platform. The ROC curve was plotted based on the corresponding NCVs for each clinical outcome. The area under the ROC curve (AUROC) was significant at 0.6896, suggesting a fair and meaningful test based on the defined cutoff and its ability to discriminate between positive and negative cases accurately (Fig 5A).

Furthermore, contingency table analysis demonstrated that of the 47 patients evaluated (includes non–QPOP-guided treatments), QPOP accurately classified 18 and 17 patients as responders and nonresponders, respectively, indicating a significant association between predicted and clinical outcomes at $P < .05$ (Fig 5B). On the basis of this stratification, clinical concordance was further verified through its sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) as well as accuracy (Fig 5B). PPV and NPV values were 69.2% and 81.0%, respectively, translating to an overall test accuracy of 74.5%, which represents the proportion of all correctly identified patients within the total patients evaluated. This demonstrates the ability of QPOP, as a CDS tool, to predict personalized therapy outcomes.

Next, to determine QPOP's utility as an FPM tool in patient treatment, we evaluated responses of 29 patients who received off-label QPOP-guided combinations (Data Supplement, Table S1). Two patients were excluded from response evaluation as no clinical assessment was conducted. Among 27 evaluable patients with R/R-NHL, the overall response rate (ORR) was 59% where seven (26%) patients achieved CR and nine (33%) achieved PR, whereas four (15%) had stable disease and seven (26%) had progressive disease (Fig 5C). Over 50% of patients achieved CR and PR within each NHL subclass (Data Supplement, Fig S5). This suggests that QPOP-guided combinations, regardless of disease subclass, had provided comparable clinical improvements.

Furthermore, to compare the overall cohort response of patients who received QPOP-guided therapies and those who did not, the ratio of PFS post-QPOP testing to PFS of treatment pre-QPOP testing (PFS_{QPOP}/PFS_{prev}) was calculated for each patient.[14] Patients who received QPOP-guided therapy (mean PFS_{QPOP}/PFS_{prev}, 3.02) had a significantly higher mean PFS_{QPOP}/PFS_{prev} ratio than those who received clinician-guided treatments (mean PFS_{QPOP}/PFS_{prev}, 1.21; Fig 5D). Additionally, approximately 59.3% of QPOP-guided patients had PFS_{QPOP}/PFS_{prev} ratios >1.3,[14] demonstrating improvements in PFS with QPOP-guided combinations compared with previous interventions (Data Supplement, Fig S6 and Table S2).

Finally, 2-year survival analysis revealed that those prescribed QPOP-guided combinations had longer median PFS than those with clinician-guided combinations (129 *v* 79 days respectively; $P = .0191$; hazard ratio (HR), 0.559 [95% CI, 0.3441 to 0.9092]; Fig 5E). Collectively, these findings demonstrate QPOP's utility as an FPM platform and highlight the overall benefit of implementing QPOP-guided treatment for R/R NHL.

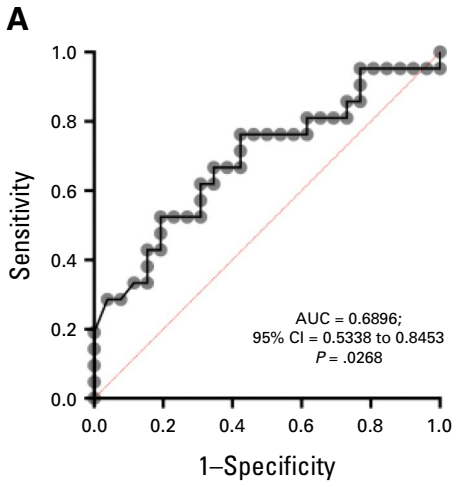
DISCUSSION

This extended, prospective study presents the clinical concordance and survival benefit of implementing QPOP as a CDS tool. Here, analytical validation of the platform using cell lines had confirmed the robustness of QPOP as a clinical laboratory developed test. More importantly, this study has highlighted the clinical concordance of QPOP and its ability to accurately predict clinical responses based on therapies identified in QPOP reports. Furthermore, patients who received QPOP-predicted combinations had an ORR of 59%. Using QPOP-predicted combinations had also significantly prolonged PFS. Our findings showcase the potential of QPOP in predicting effective, patient-specific combinations and its clinical utility as an FPM platform.

Notably, the intrapatient PFS comparison between treatment lines showed that almost 60% of QPOP-guided patients had better response durations compared with their previous treatment. This contrasts with the progressive shortening of PFS often observed with increasing lines of treatment.[10,11] Comparably, in the EXALT trial that evaluated the implementation of a single-cell, high-content microscopy-based FPM platform in advanced hematological cancers, 54% of treated patients had a 1.3-fold enhancement in PFS.[14] Similarly, in a prospective, non-interventional trial, SMARTrial, 56% of patients attained a 30% increase in PFS compared with the previous PFS.[15] The inversion of this relationship in QPOP-guided patients, coupled with the survival benefit for patients who exhausted multiple treatment options, exemplifies the clinical application of QPOP.

Additionally, we showed that QPOP effectively predicts patient treatment response, with an AUROC of 0.6896 and an accuracy of 74.5%. This is comparable with studies assessing the ability of ex vivo drug screening assays to predict clinical responses, where mean accuracy was approximately 87%.[16,35] Since off-label treatments were also evaluated, the slight dip in numbers here could be attributed to the wider repertoire of treatments used to interrogate its concordance, which contrasts with previous studies that focused on specific treatments. Nonetheless, the ability of QPOP to evaluate all possible combinations and permutations within

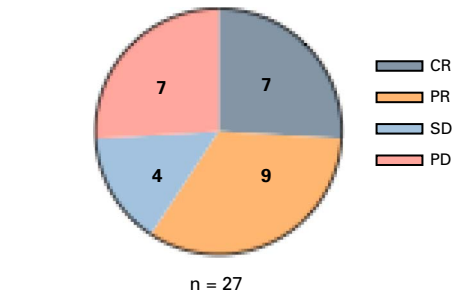
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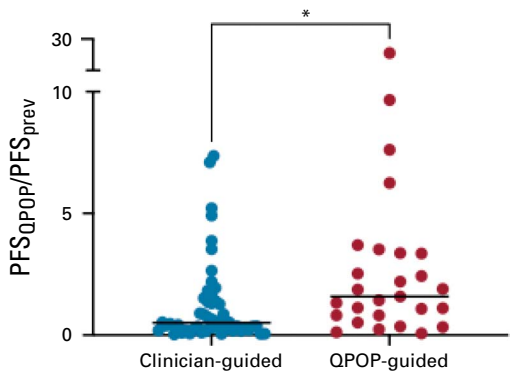


B

	Clinical Responder	Clinical Nonresponder	Total
QPOP Responder	18	8	26
QPOP Nonresponder	4	17	21
Total	22	25	$P = .001$

Sensitivity	81.8%
Specificity	68.0%
PPV	69.2%
NPV	81.0%
Accuracy	74.5%





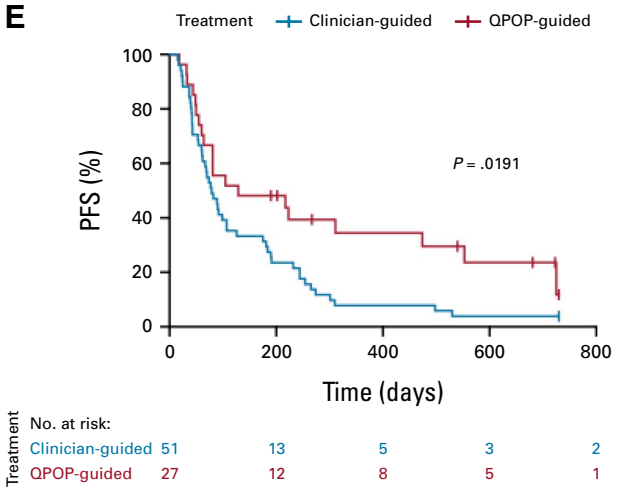


FIG 5. Evaluation of QPOP in predicting clinical responses. (A) ROC curve showing performance of QPOP in predicting response after treatment. Area under the curve, 0.6896 ($P = .0268$; 95% CI, 0.5338 to 0.8453). Statistical analysis was performed using Wilcoxon-Mann-Whitney test. (B) Contingency table of response cohorts based on defined cutoff ($P = .001$). Statistical analysis was performed using Fisher's exact test. Sensitivity, specificity, PPV, NPV, and accuracy were calculated based on the contingency table. Sensitivity = TP/(TP + FN), specificity = TN/(TN + FP), PPV = TP/(TP + FP), NPV = TN/(TN + FN), accuracy = TP + TN/total patients. (C) Number and proportions of patients with QPOP-guided, R/R-NHL patients who achieved CR, PR, SD, and PD. (D) Comparison of PFS ratio between patients who received QPOP-guided (n = 27) and clinician-guided (n = 51) treatment. PFS ratio represents the PFS of treatment received immediately after QPOP-testing (PFS_{QPOP}) over the PFS of treatment just before QPOP testing (PFS_{prev}). Statistical significance was calculated using unpaired student's *t*-test (*$P < .05$). (E) Kaplan-Meier plot displaying the 2-year survival differences between (continued on following page)

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FIG 5. (Continued). clinician- and QPOP-guided patients. Significant improvement in PFS was observed with the QPOP-guided group (P = .0191; HR, 0.559 [95% CI, 0.3441 to 0.9092]). CR, complete response; FN, false negative; FP, false positive; HR, hazard ratio; NPV, negative predictive value; PD, progressive disease; PFS, progression-free survival; PPV, positive predictive value; PR, partial response; QPOP, quadratic phenotypic optimization platform; ROC, receiver operating characteristic; R/R-NHL, relapsed/refractory non-Hodgkin's lymphoma; SD, stable disease; TN, true negative; TP, true positive.

a 12-drug search space is advantageous, particularly for ex vivo experiments with limited tissues, compared with other studies that were restricted to preselected therapies.

Extensive studies have demonstrated benefits in genomic biomarker testing for targeted cancer therapy. The National Cancer Institute-Molecular Analysis for Therapy Choice trial (ClinicalTrials.gov identifier: NCT02465060) revealed that 37.6% of patients harbored actionable mutations, with genomics-guided treatment achieving an ORR of 10.3%.[36] FPM aids to push the needle by identifying patient-specific vulnerabilities that may not be determined by genetic alterations.[12,37] Several key FPM clinical trials across different cancers have together demonstrated that FPM can be reliably used in conjunction with widely accepted genomic-based methods to guide patient treatment,[14,17,38-41] particularly those implemented with tumor board, thereby validating their clinical translation.[14,17,38,40,42] These have been explored in hematological malignancies, where treatments were initially recommended based on the ex vivo drug sensitivity results and further supplemented with genomics or transcriptomics, when available.[14,17] Recommended therapy, as decided upon the tumor board, resulted in a 59% objective response rate.[17] Furthermore, in a prospective study on R/R pediatric pan-cancers, 25% had actionable treatments based on genomic variants, whereas drug sensitivity testing identified treatment options for 100% of the samples tested.[39] With these studies highlighting the complementary effects of combining both approaches, follow-up studies should aim to integrate genomics data with the results from QPOP.

Although our study has shown promising results supporting the clinical implementation of QPOP, there are inherent limitations. Using cell lines for analytical validation may not have captured the heterogeneity of patient samples. Extending the validation to patient materials would further confirm assay robustness. Moreover, the use of a general luminescence-based cell viability readout and lack of immune cells in the QPOP platform limit the assessment of immunotherapies. Hence, interactions of immunotherapies with small molecule drugs cannot be assessed. As incorporating immunotherapy and its drug combinations is becoming increasingly popular in NHL treatment,[43,44] interrogating optimal patient-specific, immunotherapy-based drug combinations will be valuable. Since QPOP algorithm is agnostic to the nature of the phenotypic readout, future assay developments involving high-throughput co-culture of immune cells and patient-derived materials could be compatible. This would greatly augment the prediction power of QPOP, thereby expanding its clinical applications.[45]

Overall, QPOP effectively predicts patient-specific drug responses and QPOP-guided treatment especially benefitted those who failed multiple treatment lines. To further expand QPOP's application as an FPM platform, approaches to increase the number of drugs tested, incorporating tumor microenvironmental components and immunotherapies in QPOP, will be developed. Eventually, clinical implementation of QPOP and other ex vivo drug screening platforms as CDS tools will greatly improve survival chances of patients with R/R-NHL.

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